

Integrated Lifting System

Technical Datasheet - Generic Portal Reference

Document purpose: To provide a practical technical reference for integrated lifting systems used in theatre and entertainment environments. This datasheet supports portal documentation, inspection planning and site review. It does not replace manufacturer instructions, design calculations, commissioning records or competent-person assessment.

Equipment type	Integrated Lifting System
Portal category	Lifting Equipment
Typical classification	Powered lifting machinery / installed entertainment lifting system
Document status	Generic reference template - edit for manufacturer and site details
Prepared for	Stage Electrics Mechanical Theatre Equipment Compliance Portal
Revision	Draft 1.0
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Important limitation

This document is deliberately generic. The actual manufacturer, model, control architecture, load rating, travel limits, brake arrangement, duty cycle, support structure and commissioning information must be confirmed before the datasheet is used as equipment-specific technical information.

1. Equipment overview

An integrated lifting system is a complete installed lifting arrangement that may include hoists, winches, bars, control panels, drives, brakes, limits, load monitoring, emergency stops, support steelwork and associated safety devices. In theatre and entertainment environments, these systems may be used to raise, lower, suspend or position lighting bars, scenery, curtains, screens, truss or other production equipment.

- The system should be assessed as a complete lifting arrangement, not as isolated individual parts.
- The full load path should be understood from the load connection through to the supporting structure.
- Controls, limits, emergency stops, brakes, load monitoring and interlocks are safety-critical parts of the installation.
- Changes to loading, software, controls, support points or operating method should be reviewed before continued use.

- Only trained and authorised persons should operate, inspect, maintain or reset the system.

2. Equipment identification and technical details

Manufacturer / system integrator	To be confirmed
System name / model	To be confirmed
Serial number / asset ID	To be confirmed
Location / installation reference	To be confirmed
Safe Working Load / Working Load Limit	To be confirmed from design or manufacturer information
Number of lifting points / axes	To be confirmed
Drive type	To be confirmed - hoist / winch / drum / chain / wire rope / belt / other
Control system	To be confirmed - pendant / fixed panel / programmable control / group control
Safety devices	To be confirmed - limits, brakes, E-stops, load monitoring, interlocks, slack line detection
Travel range and speed	To be confirmed
Power supply and isolation	To be confirmed
Inspection interval	To be confirmed by written scheme, competent person or local inspection regime

3. Intended use

- Use only for the load types, travel ranges, speeds and operating modes intended by the manufacturer, designer or system integrator.
- Confirm the rated capacity of each lifting point and the complete system before loading.
- Ensure the load is attached using suitable accessories, suspension points and secondary arrangements where required.
- Operate the system only from the approved control position using the defined safe operating procedure.
- Confirm travel paths are clear and that persons are excluded from hazardous areas during movement.
- Do not lift people unless the complete system is specifically designed, rated and approved for that purpose.

4. Key limitations and hazards

Hazard / limitation	Why it matters	Control / check
Unverified load path	Loads may be transferred through components or structures not designed for the imposed forces.	Confirm the full load path, support points and rated capacity before use.
Control or limit failure	Movement may continue beyond safe travel or fail to stop when required.	Check limits, E-stops, control functions and safety devices in line with site procedures.
Brake or holding failure	The load may drift, descend or become uncontrolled.	Verify brake / holding function and remove from service if unreliable.
Uneven multi-point lifting	Poor synchronisation can overload one point, twist the load or damage the structure.	Confirm synchronisation, levelling, load distribution and monitoring where fitted.
Poor documentation	Users may not know capacity, operating limits, fault actions or reset procedure.	Keep manuals, drawings, commissioning records and inspection reports available.
Unauthorised modification	Software, control, structural or mechanical changes can invalidate the original safety assessment.	Record and review modifications before return to service.

5. Inspection and maintenance guidance

The following checks should be adapted to manufacturer instructions, commissioning information, the written scheme of examination and the local inspection regime. Any defect affecting lifting, braking, control, support or safety devices should be escalated and the system isolated where necessary.

Area	Typical checks	Status	Notes
Identification and rating	SWL/WLL, asset ID, system name, operating limits and warning markings are present and legible.	☐ OK ☐ Defect	
Lifting machinery	Hoists, winches, drums, chains, ropes, belts, hooks, anchors and terminations are secure and undamaged.	☐ OK ☐ Defect	
Support structure	Suspension points, beams, brackets, steelwork, fixings and anchors show	☐ OK ☐ Defect	

Area	Typical checks	Status	Notes
	no movement, cracking or corrosion.		
Controls and commands	Pendant, panel, key switch, enable controls, direction commands and indicators function correctly.	☐ OK ☐ Defect	
Safety functions	Emergency stops, upper/lower limits, overtravel protection, interlocks and load monitoring are tested where appropriate.	☐ OK ☐ Defect	
Brakes and holding	Load can be stopped and held without drift, creep, slipping or unintended movement.	☐ OK ☐ Defect	
Travel and alignment	Movement is smooth, aligned and free from abnormal noise, vibration, snatching or obstruction.	☐ OK ☐ Defect	
Electrical and control wiring	Cables, glands, connectors, enclosures, strain relief and isolation are intact and protected.	☐ OK ☐ Defect	
Documentation	Manuals, drawings, risk assessments, inspection reports, maintenance records and modification records are available.	☐ OK ☐ Defect	

6. Pre-use checks

- Confirm the system is authorised for use and has no outstanding safety-critical defects.
- Check the intended load, travel route and operating mode against the known capacity and limits.
- Check visible lifting points, bars, hooks, ropes, chains, supports and fixings before movement.
- Check that the control station, emergency stop and isolation arrangements are accessible.
- Confirm the travel path is clear of people, obstructions, cables and scenic items.
- Stop immediately if abnormal movement, noise, vibration, drift, fault indication or poor control is observed.

7. Regulatory and good-practice context

Framework	Relevance to integrated lifting systems
LOLER 1998	Relevant where the system is used for lifting or lowering loads at work. The lifting equipment and lifting operation should be suitable, marked, planned and thoroughly examined where required.
PUWER 1998	Applies as work equipment. The system must be suitable, maintained, inspected where deterioration could create risk, and used by trained people.
BS EN 17206	Highly relevant to entertainment lifting machinery. Load path integrity, control system safety, limits, brakes, documentation and prevention of uncontrolled movement should be considered.
ABTT Technical Standards	Supports theatre-industry good practice for installation, inspection, maintenance, documentation and competent operation in places of entertainment.
Competent-person note The competent person should determine the inspection and thorough examination requirements for the specific installation, taking account of manufacturer guidance, system complexity, duty, environment, frequency of use, modifications and consequences of failure.	

8. Information to retain with this datasheet

- Manufacturer manuals, operating instructions and Declaration of Conformity where applicable.
- System design drawings, load path details and structural calculations where applicable.
- Commissioning records, functional test records and safety validation documentation.
- Thorough examination reports, inspection records and maintenance history.
- Control system information, fault codes, reset procedure and emergency recovery procedure.
- Records of software, control, mechanical or structural modifications.
- Site-specific risk assessment and safe operating procedure.

9. Portal upload details

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Portal document title	Integrated Lifting System Technical Datasheet
Related equipment ID	integrated_lifting_system

10. Revision notes

Revision	Date	Change	By
1.0	08 Oct 2026	Initial generic portal datasheet.	Stage Electrics / ChatGPT support

Disclaimer: This datasheet is a guidance aid only. It must be reviewed and completed using actual manufacturer documentation, installation records, design information and competent-person findings before being relied on as equipment-specific technical data.